

Fluid Mechanics

$$\rho \triangleq \frac{\Delta m}{\Delta V} \quad SG \triangleq \frac{\rho}{\rho_{H_2O}} \quad \rho_{H_2O} = 1000[kg/m^3]$$

$$\gamma \triangleq \rho g \quad \mu [kg/m \cdot s] \quad \nu = \frac{\mu}{\rho} \text{ (kinematic viscosity) } [m^2/s]$$

$$\dot{V} = \mathbf{u}A \quad P = \dot{V}\gamma h_p \quad \dot{m} = \rho \mathbf{u}A \quad \mathbf{F}_P = - \int_A P d\mathbf{A}$$

$$\text{Newtonian:} \quad \tau = \mu \frac{\delta u}{\delta y} \quad \text{or} \quad \tau = \nu \frac{\delta u}{\delta y}$$

$$\text{Non-Newtonian:} \quad \tau \propto \frac{\delta u}{\delta y}$$

Flow Visualizations

Streamline:

$$\vec{\nabla} \times \mathbf{u} = 0 \quad \xRightarrow{z=0} \quad \frac{\delta y}{\delta x} = \frac{v}{u}$$

Pathline:

$$\begin{cases} x = x_p(t), & \frac{\delta x_p}{\delta t} = u(\mathbf{x}, t) \\ y = y_p(t), & \frac{\delta y_p}{\delta t} = v(\mathbf{y}, t) \end{cases}$$

Streakline:

Shows the history of the flow. Start at $t = 0$ and work it out step by step.

Hydrostatics

$$d\mathbf{F}_p = \vec{\nabla}(dx dy dz)$$

$$-\vec{\nabla} + \rho \mathbf{g} = 0 \quad \Rightarrow \quad \frac{\delta P}{\delta z} = -\rho g$$

$$P_{\text{bottom}} = P_{\text{top}} + \sum_{\text{down}} \gamma_i h_i - \sum_{\text{up}} \gamma_j h_j$$

$$P(z) = P_0 \left[\frac{T_0 - \alpha(z - z_0)}{T_0} \right]^{g/\alpha R}$$

Submerged Gate Procedure

1. Determine h_c from surface of fluid $\parallel \vec{g}$

2. Calculate P_c

$$P_c = \rho g h_c$$

3. Determine $\mathbf{F}_R$ acting on the gate where $y_{cp} \perp \vec{y}$

$$\mathbf{F}_R = P_c A_{\text{gate}}$$

4. Determine y_c along $\vec{y}$

5. Determine y_{cp} along $\vec{y}$

$$y_{cp} = y_c + \frac{\bar{I}}{y_c A_{\text{gate}}}$$

6. Sum torques in a useful way

$$\Sigma \tau_{\text{useful location}} = 0$$

Reynold's Transport Theorem

$$\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t} \int_{CV} b \rho dV + \int_{CS} b \rho (\mathbf{u} \cdot d\mathbf{A})$$

$$\frac{dM_{sys}}{dt} = 0 = \frac{\partial}{\partial t} \int_{CV} \rho dV + \int_{CS} \rho (\mathbf{u} \cdot d\mathbf{A})$$

$$\frac{d\mathbf{p}_{sys}}{dt} = \Sigma \mathbf{F} = \frac{\partial}{\partial t} \int_{CV} \mathbf{u} \rho dV + \int_{CS} \mathbf{u} \rho (\mathbf{u} \cdot d\mathbf{A})$$

$$\frac{d\mathbf{H}_{sys}}{dt} = \Sigma \mathbf{M} = \frac{\partial}{\partial t} \int_{CV} (\mathbf{r} \times \mathbf{u}) \rho dV + \int_{CS} (\mathbf{r} \times \mathbf{u}) \rho (\mathbf{u} \cdot d\mathbf{A})$$

Euler's Equation

$$-\frac{\partial}{\partial l}(P + \gamma z) = \rho \left(\frac{\partial \mathbf{u}_l}{\partial t} + \mathbf{u}_l \frac{\partial \mathbf{u}_l}{\partial x_l} \right) = \rho \mathbf{a}_l$$

- Incompressible
- Fluid accelerates in direction of decreasing piezo-metric head

Bernouli's Equations

$$\frac{P}{\gamma} + z + \frac{u^2}{2g} = \text{const.} \quad \text{OR} \quad \frac{P}{\gamma} + z - \frac{\omega^2 r^2}{2g} = \text{const.}$$

- Steady flow OR Solid body rotation
- Incompressible
- Applied on a streamline

Energy Equation

$$\left(\frac{P_1}{\gamma} + \alpha_1 \frac{\bar{u}_1^2}{2g} + z_1 \right) + h_p = \left(\frac{P_2}{\gamma} + \alpha_2 \frac{\bar{u}_2^2}{2g} + z_2 \right) + h_L + h_t$$

$$\alpha \triangleq \frac{\int_A \rho \frac{u^3}{2} dA}{\rho \frac{\bar{u}^3}{2} A} \quad \Rightarrow \quad \alpha = \frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right)^3 dA$$

$$\alpha = 1(\text{uniform}) \quad \alpha = 2(\text{parabolic}) \quad \alpha \approx 1(\text{turbulent})$$

Grade Lines

$$EGL = \alpha \frac{\bar{u}^2}{2g} + \frac{P}{\gamma} + z \quad HGL = \frac{P}{\gamma} + z$$

Head Loss and Friction

$$h_L = f \left(\frac{L}{D} \right) \frac{\bar{u}^2}{2g} \quad f_{\text{laminar}} = \frac{64}{Re_D}$$

$f \Rightarrow$ look at Moody diagram

Colebrook Equation

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{e/D}{3.7} + \frac{2.51}{Re^{0.9}} \right)$$

Swamee and Jain Equation

$$f = \frac{0.25}{\left[\frac{e}{3.7D} + \frac{5.74}{Re^{0.9}} \right]^2}$$

Dimensionless Terms

$$Re = \frac{\text{inertia forces}}{\text{viscous forces}} = \rho \frac{uD}{\mu} = \frac{uD}{\nu}$$

$$Eu = \frac{\text{pressure forces}}{\text{inertia forces}} = \frac{\Delta P}{\frac{1}{2} \rho u^2}$$

$$Fr = \frac{u}{\sqrt{gl}} \quad Ma = \frac{u}{c_{\text{sound}}} \quad C_D = \frac{F_D}{\frac{1}{2} \rho u^2 A}$$

Flow Similarity

1. Geometric Similarity (scale model)
2. Kinematic Similarity (streamlines/velocity field match)
3. Dynamic Similarity (non-dimensional terms are equal)

Boundary Layers

$$\delta @ u = 0.99u_0 \quad \delta^* = \int_0^\delta \left(1 - \frac{u(y)}{u_0} \right) dy$$

$$\theta = \int_0^\delta \frac{u(y)}{u_0} \left(1 - \frac{u(y)}{u_0} \right) dy$$

Laminar BL

$$\delta = \frac{5x}{\sqrt{Re_x}} \quad C_f = \frac{0.32}{Re_L^{1/7}}$$

$$\tau_{\text{wall}} = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = 0.332 \mu \frac{u_0}{x} \sqrt{Re_x}$$

Transition BL Zone

$$10^5 < Re < 10^6$$

Tripped BL

$$\delta = \frac{0.16x}{Re_x^{1/7}} \quad c_f = \frac{0.027}{Re_x^{1/7}} \quad C_f = \frac{0.032}{Re_L^{1/7}}$$

Turbulent BL

$$\delta = \frac{0.16x}{Re_x^{1/7}} \quad c_f = \frac{0.455}{\ln^2(0.06 Re_x)}$$

$$C_f = \frac{0.525}{\ln^2(0.06 Re_x)} - \frac{1520}{Re_L}$$

Drag and Lift

$$\Gamma = \oint u_l dl$$

$$\mathbf{F}_L = \int [-P \sin(\theta) - \tau \cos(\theta)] dA$$

$$\mathbf{F}_D = \int [-P \cos(\theta) + \tau \sin(\theta)] dA$$

Kutta Condition

The amount of circulation required to match reality with inviscid theory

$$\Gamma = \pi c u_0 \alpha \quad C_L = 2\pi \alpha$$

$$\mathbf{F}_L = \rho u_0^2 \pi l \alpha$$

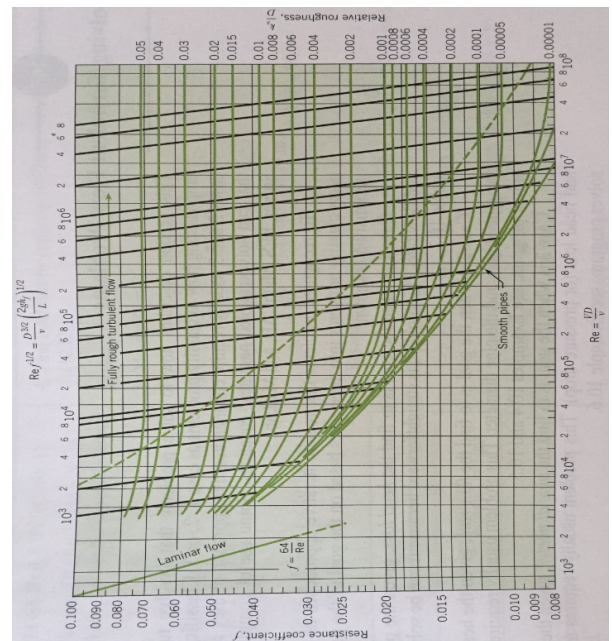


Figure 1: Moody Diagram